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Ch 5 Food and human / Ch 12.1-2 Biomolecules

Organic 有機 & Inorganic 無機

"organic products"
Food produced under the rules of organic farming 有機農業, e.g. without artificial fertilizers

Type	Definition	Examples
Organic 有機 Molecules	- Compounds with carbon, typically produced by living organisms. - include C-C & C-H bonds.	- Carbohydrates 碳水化合物, - Lipids 脂質 - Proteins 蛋白質, - Nucleic Acids 核酸 e.g. DNA - Vitamins 維生素
Inorganic 無機 Molecules	- Doesn't include C-C & C-H bonds	- Water H_2O - Minerals 礦物質 e.g. calcium 鈣, iron 鐵

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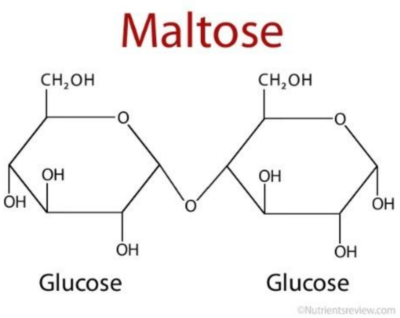
Hydrolysis & Condensation

Condensation	Molecules combine with loss of one water molecule
Hydrolysis	Molecule breakdown with use of one water molecule

Scenario	**Reaction**	**Water molecule**
Two glucoses (葡萄糖) **combine** to form one maltose (麥芽糖) 	Condensation / Hydrolysis	Gain / Loss

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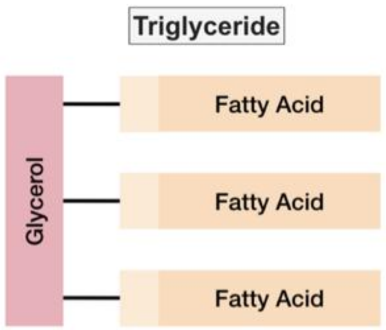
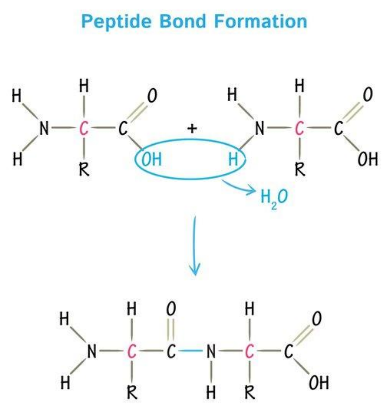
One maltose (麥芽糖) **breaks down** into two glucose molecules (葡萄糖)	Condensation / Hydrolysis	Gain / Loss
One glycerol (甘油) **combines** with three fatty acids (脂肪酸) to form a triglyceride (甘油三酯) 	Condensation / Hydrolysis	Gain / Loss
A triglyceride (甘油三酯) **breaks down** into one glycerol (甘油) and three fatty acids (脂肪酸)	Condensation / Hydrolysis	Gain / Loss
Amino acids (氨基酸) **combine** to form a dipeptide (二肽) 	Condensation / Hydrolysis	Gain / Loss

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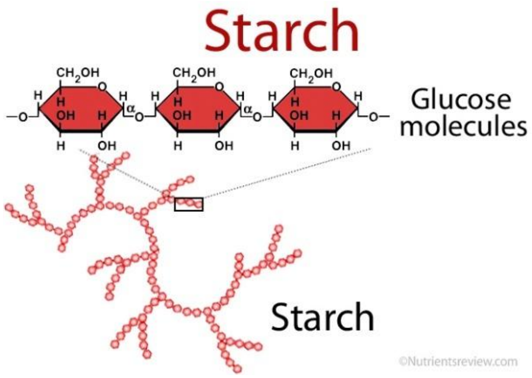
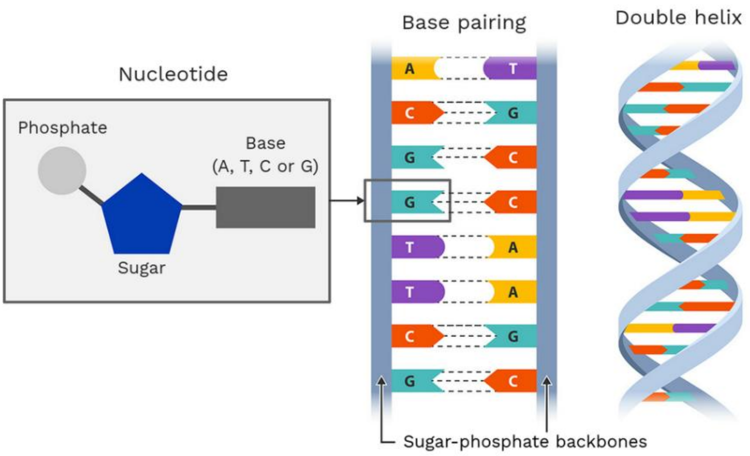
A dipeptide (二肽) **breaks down** into two amino acids (氨基酸)	Condensation / Hydrolysis	Gain / Loss
Starch (澱粉) **breaks down** into maltose (麥芽糖) molecules 	Condensation / Hydrolysis	Gain / Loss
Cellulose (纖維素) **breaks down** into glucose (葡萄糖) molecules	Condensation / Hydrolysis	Gain / Loss
DNA nucleotides (核苷酸) **combine** to form a DNA strand (DNA 鏈)	Condensation / Hydrolysis	Gain / Loss
A DNA strand (DNA 鏈) **breaks down** into nucleotides (核苷酸)	Condensation / Hydrolysis	Gain / Loss



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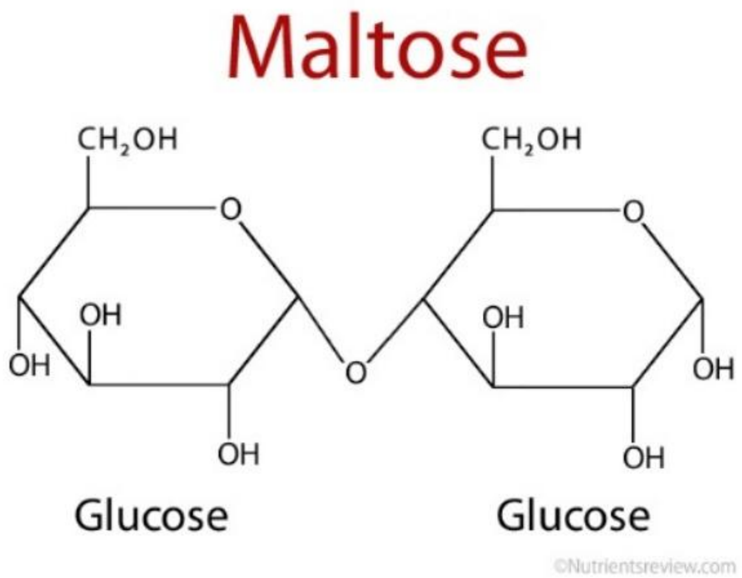
Carbohydrates 碳水化合物

example: glucose 葡萄糖 and maltose 麥芽糖



	Reaction	Number of bonds 鍵
Maltose breakdown	Condensation / Hydrolysis	

Hydrolysis reaction:

Maltose 麥芽糖 + ___ water -> glucose 葡萄糖 + glucose 葡萄糖

Condensation reaction:

glucose 葡萄糖 + glucose 葡萄糖 -> Maltose 麥芽糖 + ___ water

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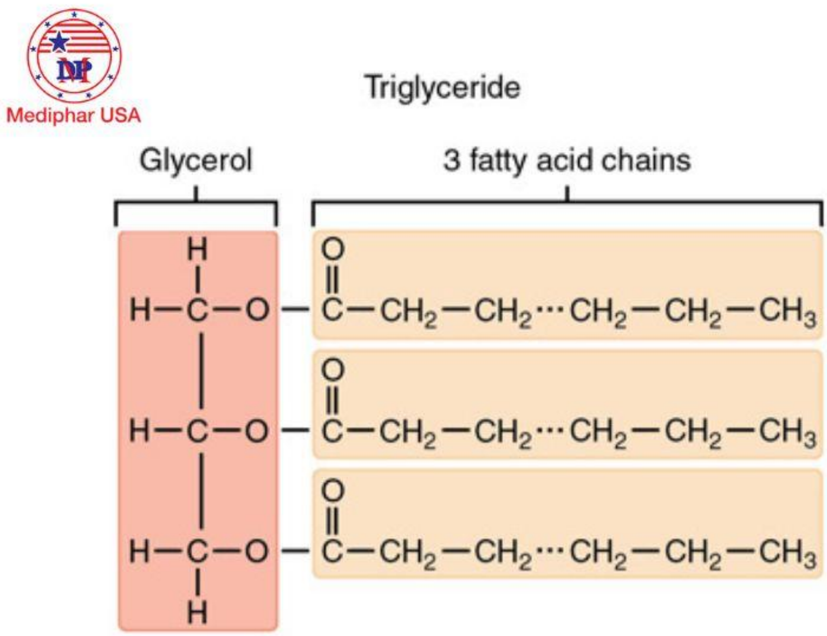
Lipids 脂質

example: triglycerides 三酸甘油酯 - fats 脂肪 and oils 油



	Reaction	Number of bonds 鍵
Triglyceride breakdown	Condensation / Hydrolysis	

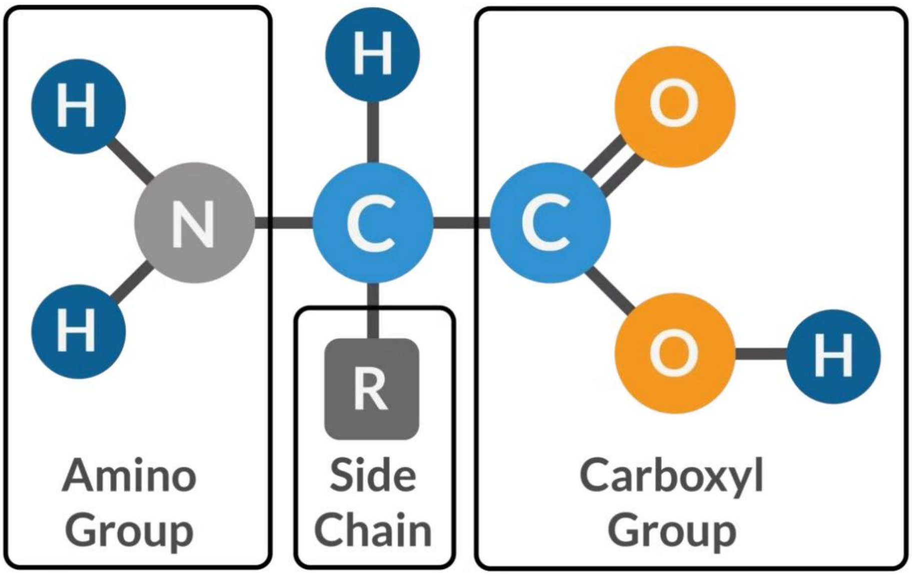
Hydrolysis reaction:

Triglyceride 甘油三酯 + __ water -> glycerol 甘油 + 3 fatty acids 脂肪酸

Condensation reaction:

Proteins 蛋白質

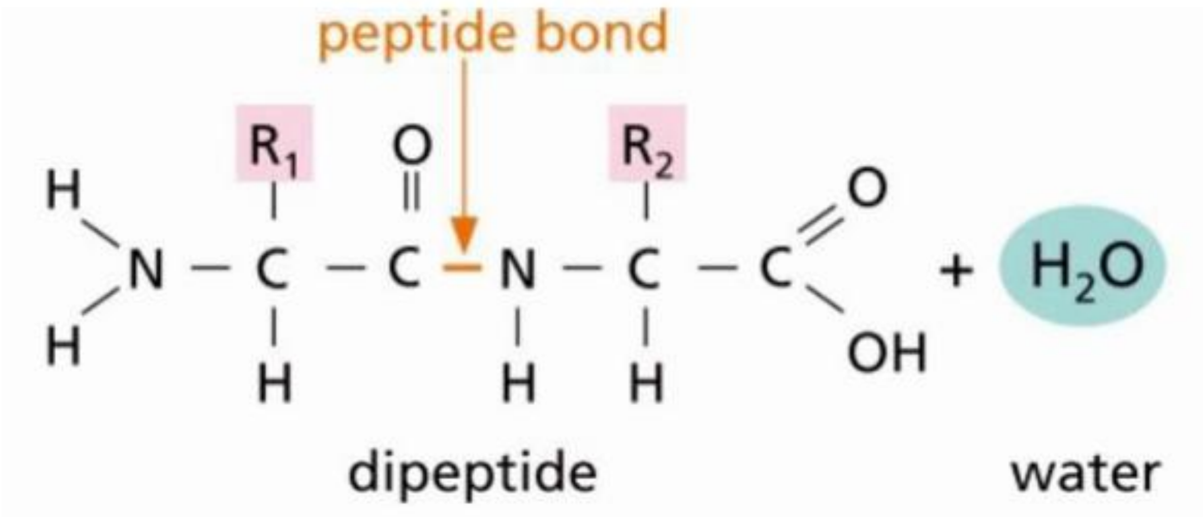
Amino acids 胺基酸 – 20 types (20 types of side chain 側鏈)



Amino acids 胺基酸

Central carbon (C), amino group (-NH2), carboxyl group(-COOH), side chain (-R)

Dipeptide 二肽



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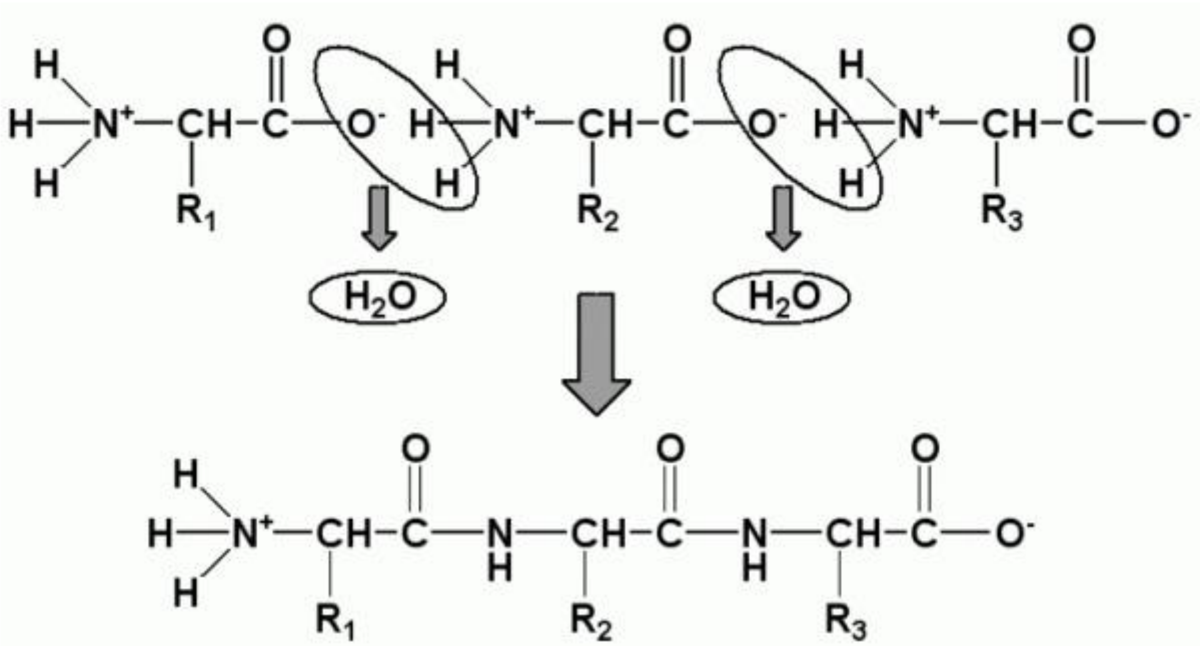
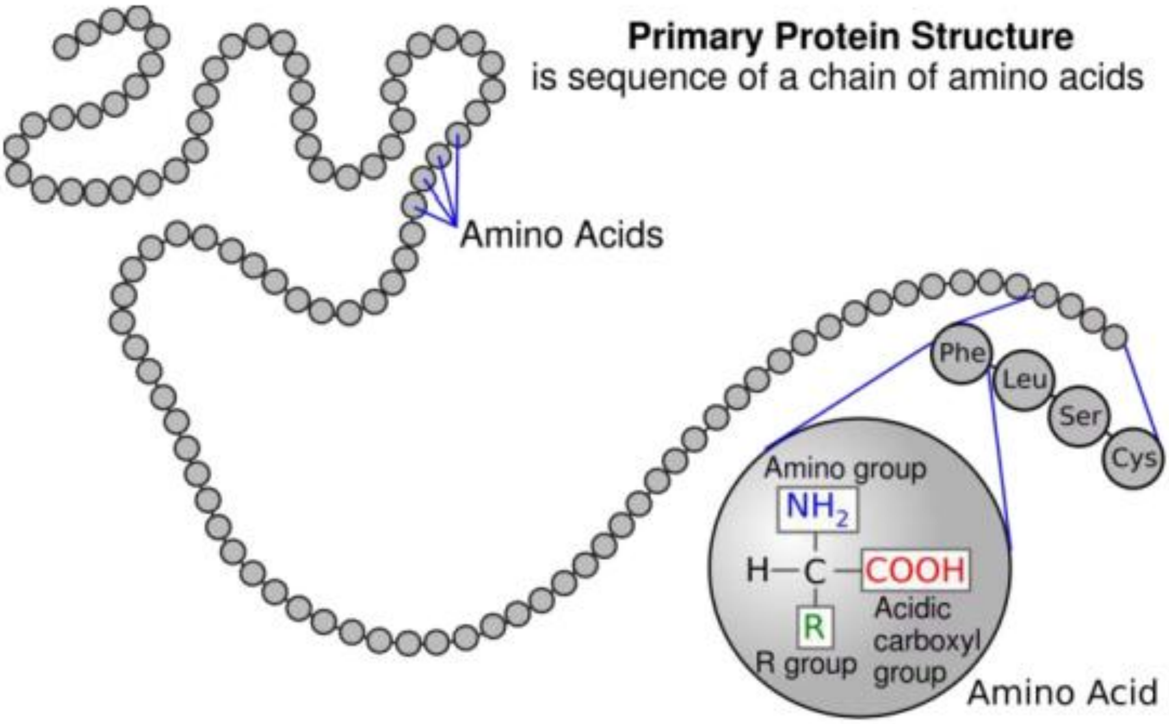
Polypeptide 多肽





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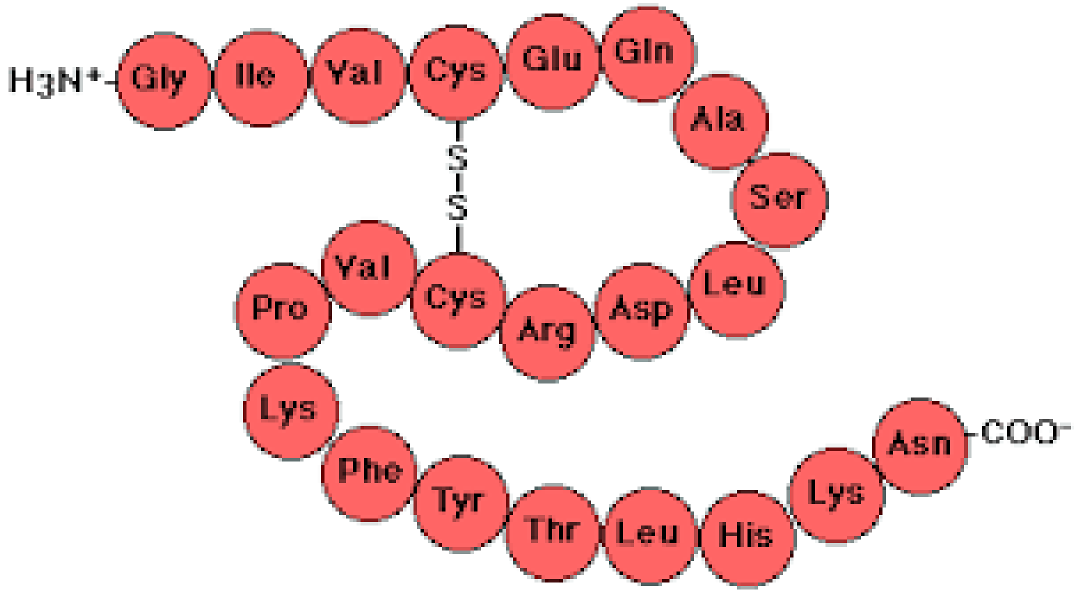
	Reaction	Number of bonds 鍵
2 amino acids 胺基酸 form 1 dipeptide 二肽	Condensation / Hydrolysis	

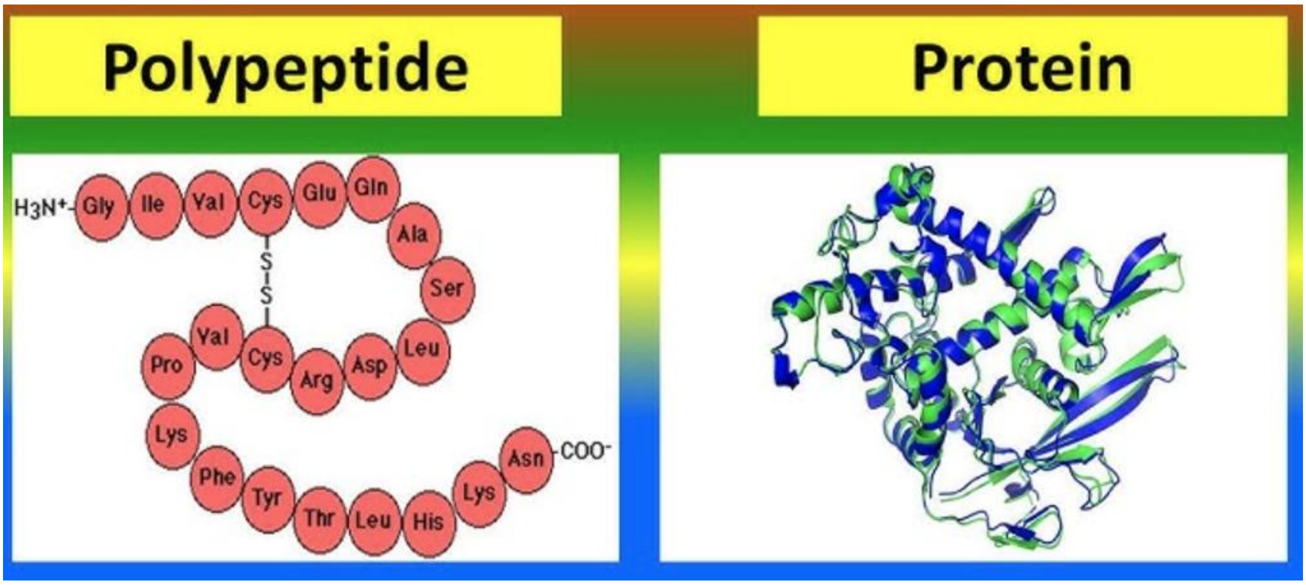
Hydrolysis reaction:

Condensation reaction:



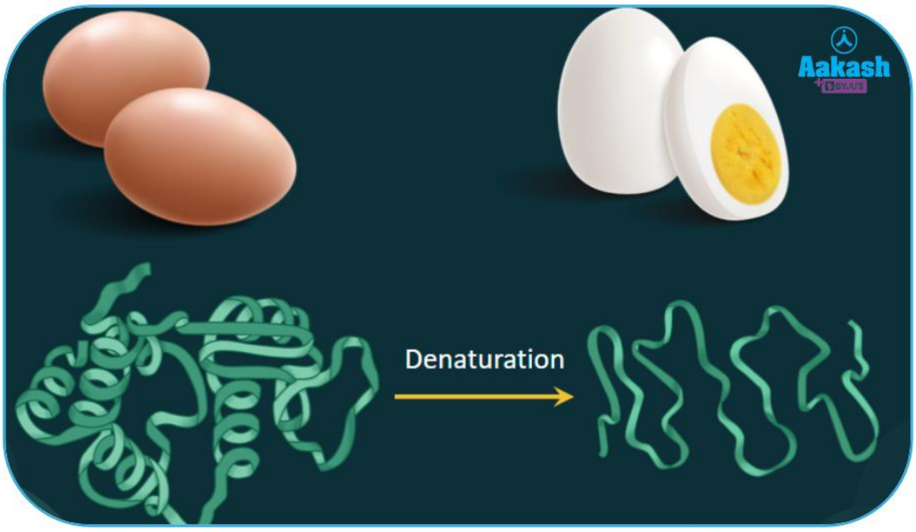
	Reaction	Number of bonds 鍵
Polypeptide 多肽 breakdown into 6 amino acids	Condensation / Hydrolysis	
5 amino acids form polypeptide	Condensation / Hydrolysis	
Polypeptide 多肽 breakdown into 22 amino acids	Condensation / Hydrolysis	

Polypeptide fold 折疊 into protein with 3D conformation 構象



Polypeptide	▶	Protein
Formed by amino acid condensation	Folding (Hydrogen bonds between amino acids)	Formed by one or more polypeptides
Linear 線性 / chain		**3D conformation** determined by **amino acid sequence** 胺基酸序列
Inactive		Specific 獨特 function

Denature 變性 is irreversible! 不可逆轉！



Transformation of Native Protein to Denatured protein

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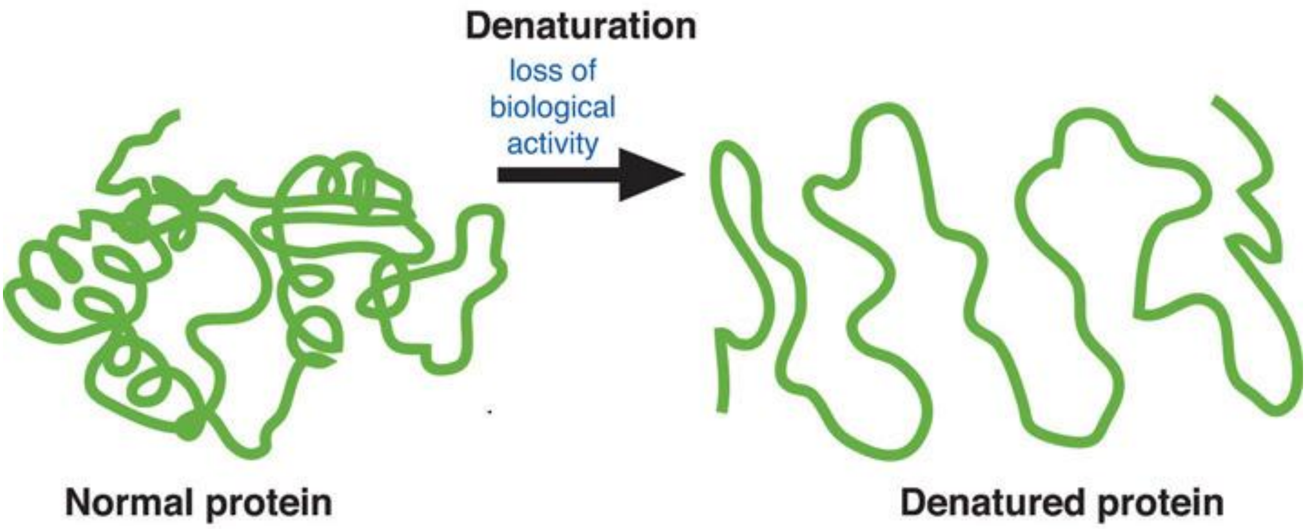
Denaturation 變性 common causes:

Non-optimal 非最優 pH, High temperature (boiling)



Denature = Loss of specific 3D conformation -> Loss of specific function

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Concept checks

Multiple-Choice Questions (MCQs)

1. Which of the following is an example of a carbohydrate?

   A. Glucose

   B. Glycerol

   C. Amino acid

   D. Triglyceride

2. Which food is rich in lipids?

   A. Rice

   B. Butter

   C. Fish

   D. Eggs

3. Which of the following processes breaks down a polypeptide into amino acids?

   A. Condensation

   B. Hydrolysis

   C. Denaturation

   D. Polymerization

4. What is the product of condensation between two amino acids?

   A. Polypeptide

   B. Dipeptide

   C. Glucose

   D. Triglyceride

5. Which of the following is a protein-rich food?

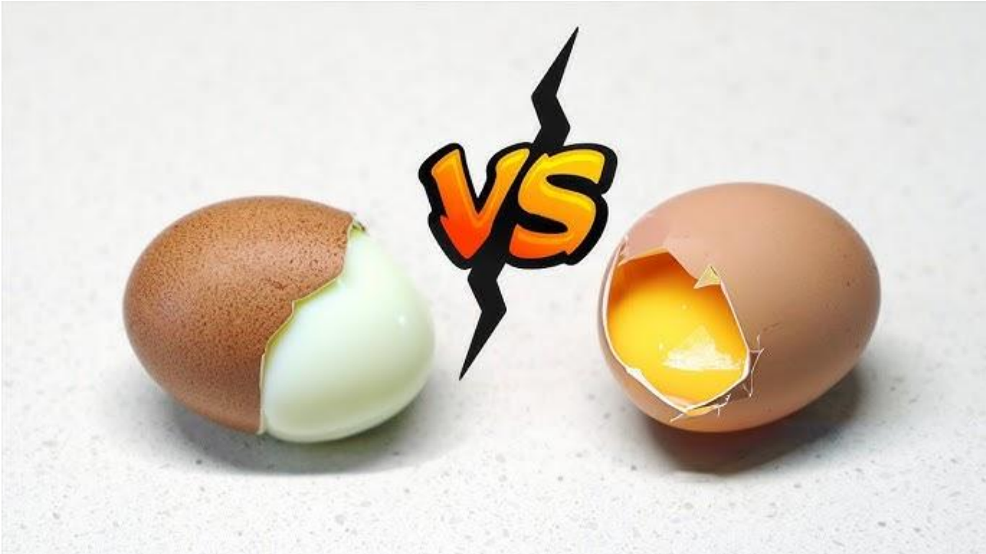
   A. Bread

   B. Cheese

   C. Avocado

   D. Honey

6. What happens during denaturation of a protein?



A. It gains 3D conformation.

B. It loses its specific shape and function.

C. It forms a dipeptide.

D. It undergoes hydrolysis.

7. Which of the following is a lipid molecule?

A. Maltose

B. Polypeptide

C. Triglyceride

D. Amino acid

8. What is the role of water in hydrolysis?

A. It creates bonds.

B. It breaks bonds.

C. It forms triglycerides.

D. It denatures proteins.

9. Which of the following foods contains carbohydrates?

A. Butter

B. Rice

C. Chicken

D. Salmon

10. Which reaction forms a triglyceride?

A. Hydrolysis

B. Condensation

C. Denaturation

D. Oxidation

11. Which of the following is a characteristic of proteins?

A. They are made of amino acids.

B. Plant-based food commonly has more proteins than animal-based food.

C. They are needed in small amounts.

D. They are simple sugars.

12. Which food is rich in carbohydrates?

    A. Milk

    B. Bread

    C. Meat

    D. Fish

13. What molecule is formed between two glucose molecules by condensation?

    A. Dipeptide

    B. Protein

    C. Starch

    D. Maltose

14. Which process breaks down a triglyceride into glycerol and fatty acids?

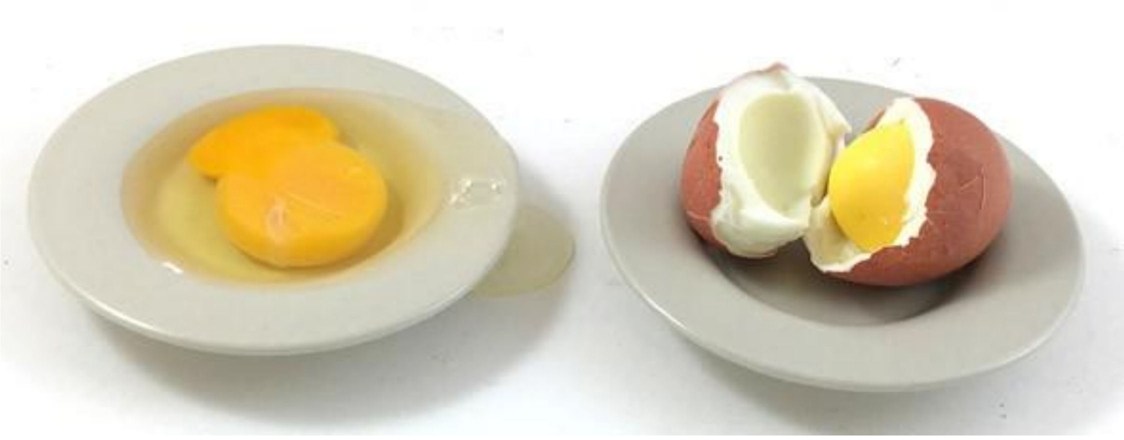
    A. Condensation

    B. Hydrolysis

    C. Denaturation

    D. Polymerization

15. What happens to a protein when it is exposed to high temperatures?



A. It forms a polypeptide.

B. It denatures.

C. It undergoes condensation.

D. It hydrolyzes.

16. Which of the following is a protein?

A. Maltose

B. Polypeptide

C. Triglyceride

D. Glucose

17. Which food is rich in protein?

A. Butter

B. Chicken

C. Potato

D. Rice

18. Which of the following foods is rich in lipids?

    A. Apple

    B. Fish

    C. Rice

    D. Bread

19. What is the main component of butter?

    A. Protein

    B. Carbohydrate

    C. Lipid

    D. Amino acid

20. Which of the following is NOT an organic molecule?

    A. Carbohydrate

    B. Protein

    C. Lipid

    D. Water

21. What is the smallest unit of a protein?

    A. Glucose

    B. Amino acid

    C. Triglyceride

    D. Polypeptide

22. Which reaction produces water as a by-product?

    A. Hydrolysis

    B. Condensation

    C. Denaturation

    D. Oxidation

23. What is the 3D structure of a protein called?

    A. Polypeptide

    B. Primary structure

    C. 3D conformation

    D. Dipeptide

24. What is the stored form of carbohydrates in plants?

A. Starch

B. Glycogen

C. Triglyceride

D. Protein

25. Which food is rich in carbohydrates?

    A. Avocado

    B. Bread

    C. Salmon

    D. Butter

26. Which of the following is an organic biomolecule?

    A. Protein

    B. Water

    C. Oxygen

    D. Carbon dioxide

27. What is the product of hydrolysis of a dipeptide?

    A. Polypeptide

    B. Amino acids

    C. Lipids

    D. Glucose

28. Which of the following is a carbohydrate?

    A. Triglyceride

    B. Maltose

    C. Polypeptide

    D. Amino acid

29. Which food is rich in proteins?

    A. Chicken breast

    B. Rice

    C. Orange

    D. Honey

30. What happens to the function of a protein when it denatures?

    A. It becomes more efficient.

    B. It loses its specific function.

    C. It forms amino acids.

    D. It produces glucose.

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True or False

Direction: put a "✓" is the sentence is correct and put a "✗" on the phrase or words that are wrong.

1. Hydrolysis breaks bonds in molecules.
2. Denaturation is a reversible process.
3. Proteins are made of amino acids.
4. Lipids are hydrophilic molecules.
5. Triglycerides are formed by condensation reactions.
6. Carbohydrates provide energy for the body.
7. Proteins are usually absent in the human body.
8. Butter is rich in carbohydrates.
9. Hydrolysis requires water to break bonds.
10. Lipids are used to form cell membranes.
11. Condensation produces water as a by-product.
12. Amino acids are the building blocks of proteins.
13. Starch is a form of stored glucose in plants.
14. Triglycerides are broken down into glycerol and fatty acids.
15. Proteins are inactive when denatured.

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Fill in the Blanks

Choose from the following:

☐ condensation ☐ hydrolysis ☐ carbohydrates ☐ proteins ☐ lipids ☐ amino acid ☐ denature ☐ triglyceride ☐ polypeptide ☐ 3D conformation

1. _________ is the process that breaks down a molecule using water.
2. _________ reactions join smaller molecules to form larger ones.
3. _________ are the main energy source for the body.
4. _________ are made up of amino acids.
5. A _________ is formed by three fatty acids and glycerol.
6. A _________ is a chain of amino acids.
7. Proteins lose their _________ when they denature.
8. _________ is the building block of proteins.
9. _________ are insoluble in water and used for energy storage.
10. A protein's specific function depends on its _________.
11. _________ reactions are required to break down triglycerides.
12. _________ is the process that forms a dipeptide from two amino acids.
13. _________ are organic molecules that include sugars and starches.
14. _________ refers to the loss of a protein's functional shape.
15. _________ are used to build tissues and enzymes in the body.

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Further details on carbohydrates, lipids and proteins

		Names	Functions	Found in
Carbohydrates	**Monosaccharides** 單醣	**Glucose 葡萄糖**	Main respiratory 呼吸作用 fuel for quick energy	Candy
		Fructose 果糖	Sweetener 甜味劑	Fruit
		Galactose 半乳糖	Component of lactose 乳糖	Milk
	Disaccharides 雙醣	**Maltose 麥芽糖**	Provides energy (提供能量)	Malt 麥芽
		Sucrose 蔗糖	Sweetener 甜味劑	Sugarcane 蔗, fruits
		Lactose 乳糖	Sugar in milk	Milk
	Polysaccharides 多醣	**Starch 澱粉**	Energy storage in plants	Potato, grains 穀物
		Cellulose 纖維素	Structural component in plant cell wall 細胞壁	Vegetables
		Glycogen 肝醣	Energy storage in animals	Liver 肝 and muscles

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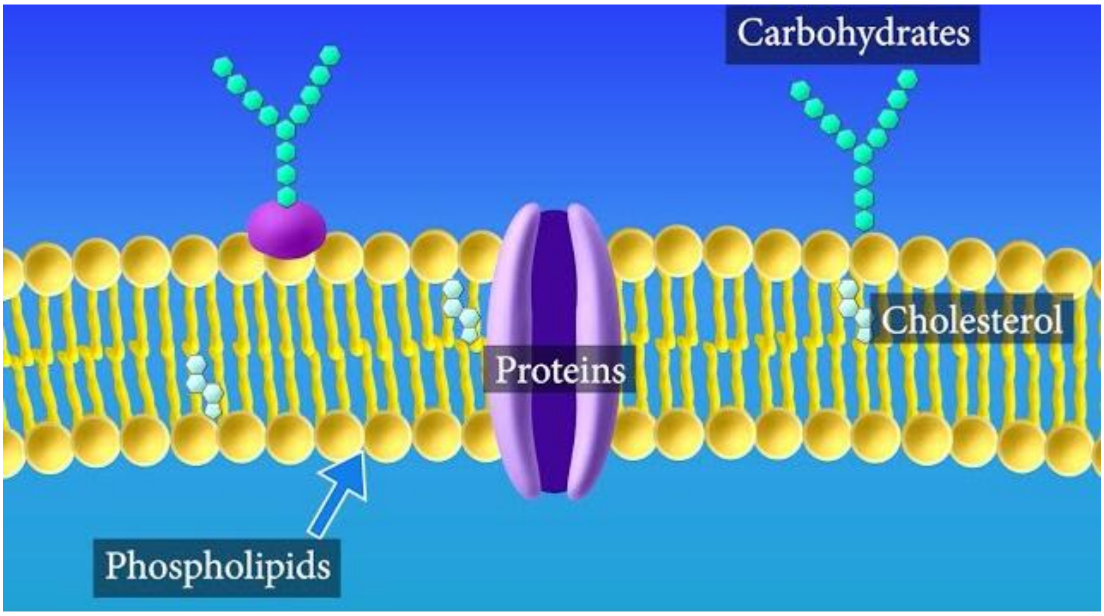

	Names		Functions	Found in
Lipids	**Triglycerides** 三酸甘油酯	**Glycerol** 甘油+ **3 fatty acids** 脂肪酸	Energy storage 能量儲存, insulation 保溫, organ protection 器官保護	Fats 脂肪 and oils 油
	Phospholipids 磷脂質	**Phosphate** 磷+ **Glycerol** 甘油+ **2 fatty acids** 脂肪酸	Main component of cell membranes 細胞膜	Cell membrane
	Steroids 類固醇	**e.g. Cholesterol** 膽固醇	Sex hormones 性荷爾蒙 cell membranes	Body tissues

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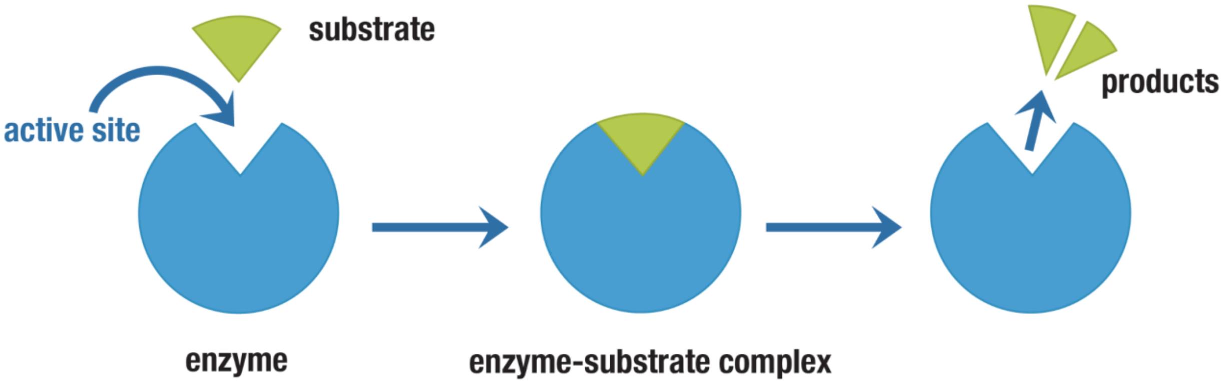

	Names	Functions	Found in
Protein	**Amino acids 胺基酸**	20 types, basic building blocks of proteins	All cells
	Dipeptide 二肽	2 amino acids	
	Polypeptide 多肽 **(Linear /chain)**	Many amino acids	
	Protein 蛋白質 **(specific 3D conformation)**	· Growth and repair 修復 of tissues 組織 · Cell membranes · Enzymes 酶: Speed up reactions · Most hormones 荷爾蒙 (in blood): Regulate body functions	Meat, eggs, beans

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Concept checks

Multiple-Choice Questions (MCQs)

1. Which of the following is a monosaccharide?

   A. Starch

   B. Glucose

   C. Lactose

   D. Sucrose

2. Which of the following is stored in the liver and muscles as an energy reserve?


   A. Cellulose

   B. Glycogen

   C. Starch

   D. Fructose

3. Which type of biomolecule is the main component of cell membranes?

   A. Proteins

   B. Lipids

   C. Carbohydrates

   D. Nucleic acids

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4. What is the main function of proteins in the human body?

A. Energy storage

B. Growth and repair of tissues

C. Insulation

D. Hormone regulation

5. Which of the following is a disaccharide?

A. Glucose

B. Maltose

C. Cellulose

D. Glycogen

6. Why should a diabetic patient 糖尿病患者 avoid consuming glucose directly?

A. It causes dehydration.

B. It raises blood sugar levels rapidly.

C. It is difficult to digest.

D. It damages the liver.

7. Which biomolecule is primarily used for quick energy in the body?

The release of energy during cellular respiration

glucose + oxygen $\xrightarrow{\text{energy released}}$ carbon dioxide + water

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$$

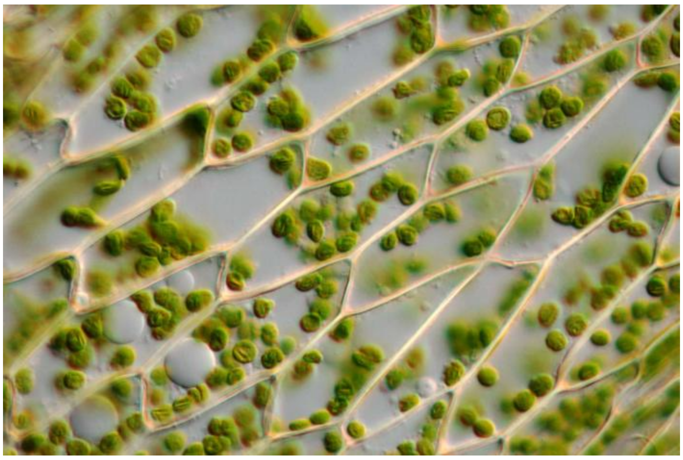
A. Proteins

B. Lipids

C. Carbohydrates

D. Steroids

8. Which of the following is a structural component in plant cell walls?



A. Glycogen

B. Cellulose

C. Starch

D. Fructose

9. What happens if a starving 飢餓 patient consumes glucose instead of bread?

A. Glucose is digested more slowly.

B. Glucose provides immediate energy.

C. Bread causes dehydration.

D. Bread is absorbed faster.

10. Which of the following lipids is essential for cell membrane structure?

A. Triglycerides

B. Phospholipids

C. Steroids

D. Fatty acids

11. Which carbohydrate is a sweetener found in fruits?

A. Glucose

B. Fructose

C. Sucrose

D. Maltose

12. What is the stored form of energy in animals?

    A. Starch

    B. Cellulose

    C. Glycogen

    D. Glucose

13. Which of the following speeds up reactions in the human body?

    A. Triglycerides

    B. Glucose

    C. Hormones

    D. Enzymes

14. What is the main energy source for respiration in humans?

    A. Proteins

    B. Glucose

    C. Lipids

    D. Vitamins

15. Which of the following is NOT a function of proteins?

    A. Growth and repair

    B. Enzyme production

    C. Energy storage

    D. Hormone regulation

16. Which part of the body stores glycogen?

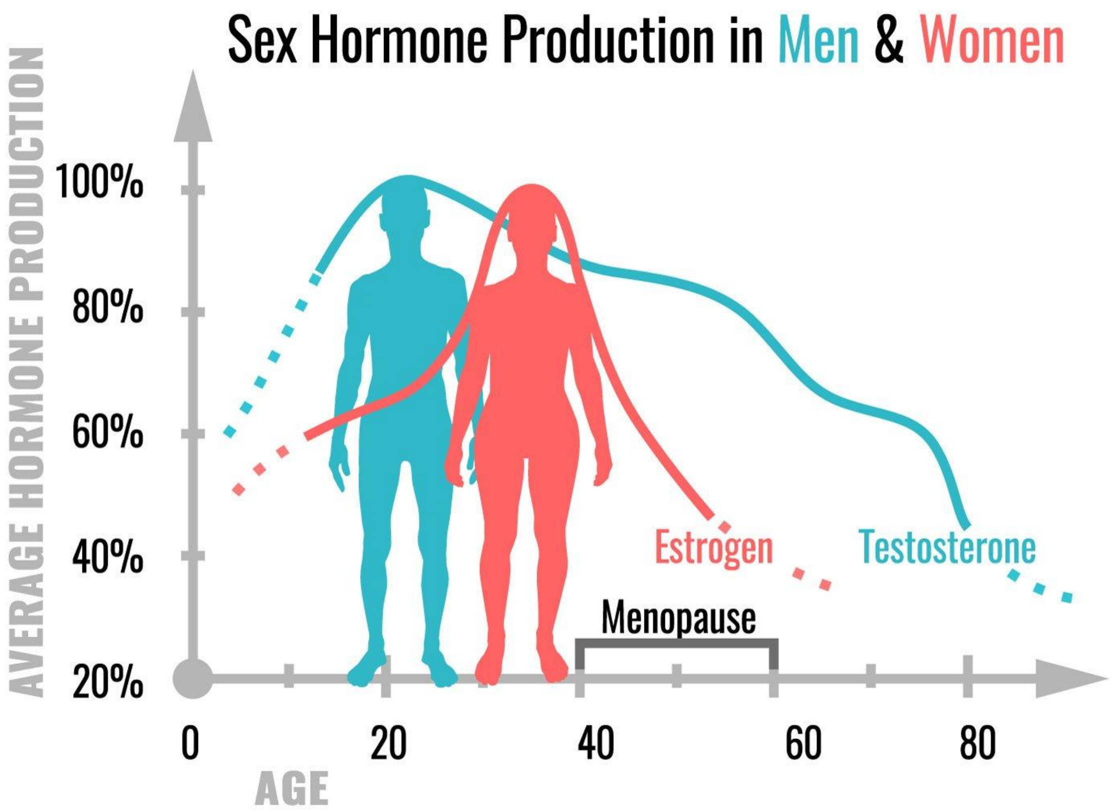
    A. Brain

    B. Liver and muscles

    C. Heart

    D. Kidneys

17. Which of the following is a lipid that regulates body functions as a hormone?



    A. Cholesterol

    B. Glycerol

    C. Phospholipid

    D. Triglyceride

18. Which carbohydrate is used as a respiratory substrate in humans?

    A. Lactose

    B. Starch

    C. Glucose

    D. Cellulose

19. Why is milk a good source of energy for infants?

    A. It contains proteins only.

    B. It contains lactose, a disaccharide.

    C. It contains starch.

    D. It contains cellulose.

20. Why do athletes consume food rich in starch before a match?

    A. To build muscle mass.

    B. To store energy for long-term use.

    C. To provide glucose steadily for energy during the race.

    D. To repair tissues.

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True or False

Direction: put a “✓” is the sentence is correct and put a “✗” on the phrase or words that are wrong.

1. Glucose is a monosaccharide.
2. Proteins are used for energy storage in the body.
3. Cellulose is found in the cell walls of plants.
4. Glycogen is stored in the liver and muscles.
5. Lipids are the main source of quick energy in humans.
6. Enzymes are proteins that speed up chemical reactions.
7. Fructose is a disaccharide found in fruits.
8. Starch is a storage carbohydrate in plants.
9. Phospholipids are the main components of cell membranes.
10. Proteins are required for growth and repair in humans.
11. Steroids are a type of lipid that acts as a hormone.
12. Maltose is a monosaccharide.
13. Glycogen is the stored form of glucose in plants.
14. Non-green parts of plants, like potatoes, store starch.
15. Lactose is a carbohydrate found in milk.

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Fill in the Blanks

Choose from the following:

☐ glycogen ☐ starch ☐ glucose ☐ enzymes ☐ proteins ☐ fructose ☐ lactose ☐ triglycerides ☐ phospholipids ☐ cellulose

The main carbohydrate stored in animals is _________.

1. _________ is the carbohydrate stored in plant cell walls.
2. Milk contains _________, a disaccharide.
3. _________ are used to speed up chemical reactions in the body.
4. The stored form of glucose in plants is _________.
5. _________ is a monosaccharide that provides quick energy.
6. The main lipid component of cell membranes is _________.
7. _________ is a carbohydrate found in fruits.
8. Proteins are essential for _________ and repair of tissues.
9. _________ is a lipid used for energy storage and insulation.
10. _________ is a carbohydrate used by athletes for quick energy.
11. _________ is the storage carbohydrate in the liver and muscles.
12. _________ is the carbohydrate found in milk.
13. _________ are the building blocks of proteins.
14. _________ is the structural carbohydrate found in plant cell walls.

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Answer Key

Concept check – basics of carbohydrates, proteins and lipids

MCQs:

1. A 2. B 3. B 4. B 5. B 6. B 7. C 8. B 9. B 10. B
2. A 12. B 13. D 14. B 15. B 16. B 17. B 18. B 19. C 20. D
3. B 22. B 23. C 24. A 25. B 26. A 27. B 28. B 29. A 30. B

True/False:

1. ✓ 2. ✗ (Denaturation is irreversible) 3. ✓ 4. ✗ (Lipids are hydrophobic)
2. ✓ 6. ✓ 7. ✗ 8. ✗ (Butter is rich in lipids)
3. ✓ 10. ✓ 11. ✓ 12. ✓ 13. ✓ 14. ✓ 15. ✓

Fill in the Blanks:

1. Hydrolysis
2. Condensation
3. Carbohydrates
4. Proteins
5. Triglyceride
6. Polypeptide
7. 3D conformation
8. Amino acid
9. Lipids
10. 3D conformation
11. Hydrolysis
12. Condensation
13. Carbohydrates
14. Denature
15. Proteins

Concept check – Functions and names of carbohydrates, proteins and lipids

MCQs

1. B 2. B 3. B 4. B 5. B
2. B 7. C 8. B 9. B 10. B
3. B 12. C 13. D 14. B 15. C
4. B 17. A 18. C 19. B 20. C

True/False

1. ✓ 2. ✗ (Respiration occurs all the time.)
2. ✓ 4. ✓ 5. ✓
3. ✓ 7. ✗ (Fructose is a monosaccharide.)
4. ✓ 9. ✓ 10. ✓

5. ✓ 12. ✗ (Maltose is a disaccharide.)
6. ✗ (Glycogen is stored in animals.)
7. ✓ 15. ✓

Fill in the Blanks

1. Glycogen
2. Cellulose
3. Lactose
4. Enzymes
5. Starch
6. Glucose
7. Phospholipids
8. Fructose
9. Growth
10. Triglycerides
11. Glucose
12. Glycogen
13. Lactose
14. Proteins
15. Cellulose